

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Computer Engineering/Computer Science & Engineering

Semester : VI

Subject Code & Name: BTCOE604B_Y23 Internet of Things

Max Marks: 60

Date: 26/12/24

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The IoT started between the years _____ a. 2008-2009 b. 2002-2003 c. 2004-2005 d. 2018-2019	Remember	1
2	Self-driving car is now becoming a reality with well-known projects like _____ self-driving car. a. Window's b. Google's c. Amazon d. None of these	Analysis	1
3	IP, _____, and UDP bring connectivity to IoT networks. a. TCP b. Internet c. UDP d. IP	Understand	1
4	_____ are fundamental building blocks of IoT networks a. IP b. Sensors c. IOT d. a &b	Remember	1
5	_____ is a wireless access technology for low-cost and low-data-rate devices a. IEEE 802.15.4 b. IEEE 802.2.1 c. IEEE 802.1.2 d. None of these	Understand	1
6	IEEE 802.11 _____ is certainly the most successfully deployed wireless technology. a. Wi-Fi b. LAN c. MAN d. WAN	Analysis	1
7	Routing Protocol for Low Power and Lossy Networks a. IPV4 b. ICMP c. IPV6 d. UDP	Understand	1
8	_____ is one of the most common industrial protocols in the world a. SCADA b. Wi-Fi c. 3G d. 4G	Analysis	1
9	In _____ learning, the machine is trained with input for which there is a known correct answer. a. supervised b. unsupervised c. training set d. pixel	Understand	1
10	Smart city IoT infrastructure is a _____ architecture. a. four-layered b. five-layered c. three-layered d. two-layered	Remember	1

11	In order to promote _____ the government should employ more management.		Analysis	1
	a. eGovernment related	b. Enterprise-based	c. Company based	d. Business oriented platform
12	A _____ is thermally sensitive resistor that exhibits a large change in resistance.		Synthesis	1
	a. Thermistor	b. Thermocouple	c. LVDT	d. Pressure sensor
Q. 2 Solve the following.				
A)	What Is IOT? Explain Genesis of IOT.		Knowledge	6
B)	Explain IOT data management and computing stack.		Understand	6
Q.3 Solve the following.				
A)	Explain how sensor and actuator interact with the physical world.		Understand	6
B)	Explain Communication Protocols for Wireless Sensor Networks.		Remember	6
Q. 4 Solve Any Two of the following.				
A)	Explain Optimizing IP for IoT Using an Adaptation Layer.		Understand	6
B)	Comparison of an IoT Protocol Stack Utilizing 6LoWPAN and an IP Protocol Stack.		Analysis	6
C)	Explain CoAP Communications in IoT Infrastructures.		Remember	6
Q.5 Solve Any Two of the following.				
A)	Explain Message Queuing Telemetry Transport (MQTT).		Understand	6
B)	Explain IoT Application Layer Protocols.		Remember	6
C)	Explain Massively Parallel Processing Databases.		Understand	6
Q. 6 Solve Any Two of the following.				
A)	Explain Smart City Security Architecture with a neat diagram.		Understand	6
B)	Explain different categories of Raspberry operating system.		Remember	6
C)	Explain the Data types, operators and constants in Arduino programming.		Synthesis	6
*** End ***				